

PAPER 1 / QUESTION 172 May 2023 / TZ1 (file label) / HL Paper 1 / original question 5 / 2223-6507

- 17.** A car accelerates uniformly. The car passes point X at time t_1 with velocity v_1 and point Y at time t_2 with velocity v_2 . The distance XY is s .



The following expressions are proposed for the magnitude of its acceleration a :

I.
$$a = \frac{2s}{(t_2 - t_1)^2}$$

II.
$$a = \frac{v_2^2 - v_1^2}{2s}$$

III.
$$a = \frac{v_2 - v_1}{t_2 - t_1}$$

Which is correct?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III